

Exp No: 12

EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS

OBJECTIVES

- To write an Embedded C program to multiply two 16-bit binary numbers using Keil software.
- To obtain and store the resulting 32-bit product.
- To display the 32-bit result through four 8-bit ports (P0, P1, P2, and P3).

PROGRAM

```
#include <reg51.h> void main()
{
while (1)
{
unsigned int num1, num2; unsigned long int product; num1 = 0x2222;
num2 = 0xBBBB;
product = (unsigned long int)num1 * num2; P0= product &0xff;
P1= (product & 0xff00)>>8;
P2 =(product & 0xff0000)>>16; P3 =(product & 0xff000000)>>24;
}
}
```

OUTPUT

The screenshot displays the uVision IDE interface with a C program being simulated. The program calculates the product of two numbers and outputs the result to four parallel ports.

Registers:

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x19
Sys	
a	0x19
b	0x18
sp	0x08
sp_max	0x11
dptr	0x0000
PC	0x0800
status	1241
sec	0.00062050
paw	0x01

Disassembly:

```
2: {
3:   while (1)
4:   {
5:     unsigned int num1, num2; unsigned long int product; num1 = 0x2222;
6:     num2 = 0xB88B;
7:     product = (unsigned long int)num1 * num2; P0= product & 0xff;
8:     P1= (product & 0xff00)>>8;
9:     P2 =(product & 0xff0000)>>16; P3 =(product & 0xff000000)>>24;
10:  }
11: }
```

Parallel Port 0:

Port 0	7	Bits	0
P0	0x06	✓	✓
Pins	0x06	✓	✓

Parallel Port 1:

Port 1	7	Bits	0
P1	0xC4	✓	✓
Pins	0xC4	✓	✓

Parallel Port 2:

Port 2	7	Bits	0
P2	0x07	✓	✓
Pins	0x07	✓	✓

Parallel Port 3:

Port 3	7	Bits	0
P3	0x19	✓	✓
Pins	0x19	✓	✓

Command:

```
Running with Code Size Limit: 2K
Load "C:\\Users\\M.POOJITH GANESH\\OneDrive\\embedded\\Experiment12\\Objects\\Exp12"
```

Watch 1:

Name	Value	Type
num1	0x2222	uint
num2	0xB88B	uint
product	0x1907C4D6	ulong
P0	0x06	uchar
P1	0xC4	uchar
P2	0x07	uchar

Simulation: t1: 0.00062050 sec L5 C1 CAP: NUM SCRL: OVR: R/W